AI-Powered Customer Experience Transformation Report

Meridian Retail Group | Strategic Technology Division
Prepared by: Office of the Chief Digital Officer | June 2026

1. Executive Summary

Meridian Retail Group (MRG) operates 214 retail outlets across 18 states and processes approximately 2.3 million customer transactions per month through its digital and in-store channels. Following a 12% year-over-year decline in customer satisfaction scores (CSAT) and a 19% increase in support ticket volume in FY 2025, the board of directors approved a $4.2 million AI Transformation Initiative in Q1 2026.

This report outlines the recommended architecture, vendor selections, cost projections, risk assessment, and 18-month implementation roadmap for deploying AI across three core business functions: Customer Support Automation, Personalised Marketing, and Supply Chain Demand Forecasting.

Key projected outcomes at 18-month mark: 42% reduction in average support resolution time, 28% improvement in CSAT scores, $1.8M annual savings in customer operations, and a projected 11% uplift in repeat-purchase revenue. 3-year NPV: $9.4M. Break-even modelled at month 17.

2. Problem Statement & Business Context

2.1 Current Operational Pain Points

- Support team handles 47,000 tickets/month; 61% are repetitive (order status, returns, billing).
- Average first-response time is 6.4 hours; industry benchmark is 1.2 hours.
- Customer churn rate increased from 8.1% to 11.4% between Q3 2024 and Q1 2026.
- Marketing email open rates dropped to 14.2%, below the 21% industry average.
- Stockout events cost an estimated $3.1M in lost revenue in FY 2025.

2.2 Strategic Objectives & KPIs

The initiative directly supports Pillar 3 (Digital-First Customer Experience) and Pillar 5 (Operational Efficiency) of MRG's five-year strategic plan. The CDO office has set these non-negotiable KPIs:

- CSAT score: improve from 3.1/5 to 4.0/5 by December 2026.
- Support cost per ticket: reduce from $18.40 to under $9.00.
- First-contact resolution rate: increase from 34% to 65%.

- Marketing email conversion rate: improve by at least 15% via personalisation.
- Stockout incidents: reduce from 142/month to fewer than 57/month.

3. Recommended AI Architecture

3.1 Pillar A — Intelligent Customer Support

We recommend a RAG-based (Retrieval-Augmented Generation) support assistant built on Google Gemini 2.5 Flash as the primary LLM, with LangChain as the orchestration layer. Customer queries are routed through an intent classifier before reaching the RAG pipeline. Tier-1 queries (order status, FAQs) are resolved autonomously; Tier-2 (complaints, refunds) are escalated to human agents with an AI-generated summary and suggested reply.

Vector store: Pinecone (us-east-1 cluster) indexed against MRG's 1,200-page product knowledge base, return policies, and 480,000 historical resolved tickets. Estimated index size: 2.1M vectors at 768 dimensions.

3.2 Pillar B — Personalised Marketing Engine

A collaborative filtering model trained on 14 months of purchase history (3.8M transactions) will power product recommendations in email campaigns and the mobile app. LLM-generated personalised copy replaces static email templates. Expected to reduce unsubscribe rates by 22% and increase average order value (AOV) by $12.50.

3.3 Pillar C — Demand Forecasting

A Prophet + XGBoost ensemble model integrated into the ERP system will predict SKU-level demand 6 weeks in advance across all 214 stores. Training data: 5 years of sales history, weather data, and promotional calendar. Target: reduce stockouts by 60% and overstock write-offs by 35%.

4. AI Vendor Evaluation & Selection

The evaluation committee assessed six platforms across five criteria: generation quality, cost at scale, data privacy compliance, ease of integration with Salesforce and SAP, and enterprise SLA availability.

4.1 Selected Vendors

Primary LLM:	Google Gemini 2.5 Flash — best cost/quality ratio at our query volume; 1M token context window; GDPR compliant via Google Cloud EU region.
Embedding Model:	Gemini Embedding-001 — 768-dim vectors; outperformed OpenAI text-embedding-3-small by 4.1% on our internal retrieval benchmark.
Vector Database:	Pinecone (Production) / FAISS (Development) — Pinecone chosen for 99.99% uptime SLA; FAISS used in dev/staging to eliminate cost.
Orchestration:	LangChain 1.x LCEL pipeline — avoids vendor lock-in.

MLOps:

	Google Vertex AI Pipelines — auto retraining on drift > 0.08.

4.2 Rejected Alternatives

- OpenAI GPT-4o: 2.3x higher cost at 180,000 queries/month; US-only data residency creates GDPR complications for EU customer data.
- AWS Bedrock (Claude 3): 14-week procurement vs 3 weeks for GCP; high integration complexity with non-AWS stack.
- Ollama (local models): Insufficient accuracy on intent classification; $220K GPU CapEx not approved in current budget cycle.

5. Budget Allocation & Cost Analysis

5.1 Total Approved Budget: $4,200,000

Implementation & Integration (Year 1):	$1,850,000
LangChain/API development (internal):	$620,000
Pinecone enterprise contract (3-year):	$180,000
Google Cloud AI APIs (Year 1):	$340,000
Data engineering & ETL pipelines:	$410,000
Security audit & compliance:	$300,000
Ongoing Annual Operating Cost (Year 2+):	$890,000
Gemini API calls ($0.0003/1K tokens):	$216,000
Pinecone annual renewal:	$60,000
Monitoring, maintenance, retraining:	$190,000
Human oversight team (3 FTE):	$424,000

Projected ROI: $1.8M support savings + $2.1M personalisation uplift + $3.1M stockout reduction = projected 3-year NPV of $9.4M on a $6.87M total investment. Break-even at month 17.

6. Risk Assessment & Mitigation Strategies

Risk: LLM Hallucination in Support Responses

Likelihood:	HIGH
Business Impact:	MEDIUM
Mitigation:	RAG grounding mandatory; human-in-the-loop for Tier-2; judge model scores factuality before delivery.

Risk: Customer Data Privacy (GDPR/CCPA)

Likelihood:	HIGH
Business Impact:	LOW
Mitigation:	PII stripped before LLM calls; EU GCP data residency; DPA signed with Google Cloud; annual privacy audit.

Risk: Model Drift Over Time

Likelihood:	MEDIUM
Business Impact:	MEDIUM
Mitigation:	Automated drift detection; quarterly retraining; fallback to rule-based if accuracy drops below 78%.

Risk: Vendor Lock-in (Google Cloud)

Likelihood:	MEDIUM
Business Impact:	LOW
Mitigation:	LangChain abstraction enables swap-out; Anthropic Claude pre-evaluated as fallback vendor.

Risk: Staff Resistance

Likelihood:	MEDIUM
Business Impact:	LOW
Mitigation:	8-week change management; AI framed as augmentation; $85,000 reskilling budget allocated.

Risk: SAP ERP Integration Failures

Likelihood:	LOW
Business Impact:	HIGH
Mitigation:	3-month dedicated integration sprint; SAP consultant contracted at $120K for Pillar C.

7. 18-Month Implementation Roadmap

Phase 1 — Foundation (July – September 2026)

- Deploy RAG FAQ assistant on web chat; target 30% Tier-1 ticket deflection.
- Index full 1,200-document knowledge base into Pinecone.
- Set up Grafana monitoring; establish CSAT baselines.
- Milestone: First live AI support response by 1 August 2026.

Phase 2 — Expansion (October – December 2026)

- Extend AI support to email and WhatsApp channels.
- Launch personalised recommendations for top 500K customers.
- Begin demand forecasting model training on 3 years of sales history.
- Milestone: CSAT reaches 3.6/5; cost per ticket below $14.

Phase 3 — Full Deployment (January – December 2027)

- Demand forecasting live across all 214 stores; ERP integration complete.
- AI recommendations active for all 2.1M registered customers.
- AI writes and A/B tests all promotional email campaigns.
- Milestone: All KPIs from Section 2.2 achieved by December 2027.

8. Governance & Success Metrics

8.1 Steering Committee

Executive Sponsor:	Ms. Priya Nair, Chief Digital Officer
Programme Director:	Mr. James Okafor, VP of Technology
AI Ethics Lead:	Dr. Lena Hoffmann, Head of Data & Privacy
Delivery Partner:	InnovateTech Consulting Ltd. (ref: MRG-2026-0042)

8.2 Quarterly Review Cadence

Steering committee convenes first Monday of each quarter. A go/no-go gate is scheduled 30 September 2026 before Phase 2 funding is released. Independent QA review by KPMG scheduled for Q2 2027.

8.3 Monthly KPI Dashboard

Metric	Current Baseline	Target
Tickets resolved autonomously	0%	55%
Avg. first-response time	6.4 hours	< 45 min
CSAT score	3.1 / 5.0	4.0 / 5.0
Support cost per ticket	$18.40	< $9.00
Email click-through rate	3.8%	6.5%
Stockout incidents / month	142	< 57
Repeat-purchase revenue uplift	Baseline	+ 11%